

An Accelerated Approach to Time-Sequential Production Simulation for Integrated Generation-Grid-Storage Systems Combining Benders Decomposition and Data-Driven Methods

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Abstract—The rapid growth of wind and solar power has made electricity system simulations increasingly complex, with high-dimensional, tightly interconnected variables that overwhelm traditional solution methods. To address this challenge, this paper proposes a new approach that combines Benders decomposition with data-driven techniques to accelerate large-scale, time-based simulations of integrated generation, grid, and storage operations under high renewable penetration. The method begins with a detailed analysis of the simulation model, which is mathematically structured as a large-scale mixed-integer linear program. Using decomposition, the model is split into a master problem dealing with discrete decisions and subproblems focused on continuous variables. To further streamline computation, the framework incorporates a pre-screening mechanism based on Gaussian process models. This mechanism uses patterns from past solutions to identify and remove inactive parts of the feasible region and redundant safety constraints, effectively narrowing the search space. A case study involving a 90-node provincial power system shows that the proposed method significantly improves computational efficiency without compromising accuracy.

Keywords—Coordinated Time-Series Simulation of Generation-Grid-Storage Systems, High-Dimensional Mixed-Integer Linear Programming, Benders Decomposition, Gaussian Process

I. INTRODUCTION

As China presses ahead with its “dual carbon” goals, its power system is undergoing a rapid transformation toward a cleaner, low-carbon structure—one built on multi-energy integration and closer coordination between supply and demand. The fast-paced expansion of renewable energy has made managing the grid more complex. Wind and solar power, with their inherent variability and uncertainty, can cause issues like frequency instability and voltage deviations. At the same time, the different parts of the system—generation, transmission, consumption, and storage—are becoming more tightly interconnected across both time and space, amplifying risks such as power flow disruptions and imbalances between supply and demand^[1-3]. These challenges highlight the urgent need for efficient tools that can assess energy balance risks and support the safe and cost-effective operation of the power system.

As renewables take up a growing share of the energy mix, grid operations have become increasingly fast-changing and dynamic. To maintain system balance, conventional power plants are now required to ramp up and down more frequently. Time-sequential production simulation—an approach that captures the temporal dynamics of generation and demand

using hourly data—has emerged as a key tool for optimizing medium- to long-term operations in high-renewable power systems. One study [4] used this method to optimize the configuration of wind, solar, thermal, and storage systems in a direct current near-field setting. Another [5] proposed a multi-timescale simulation-based approach to assess power balance risk. A separate study [6] developed a new frequency risk assessment framework grounded in time-sequential simulation, offering robust support for the secure and stable operation of real-world power systems.

Time-sequential production simulation must account for the operational constraints of multiple types of equipment—spanning generation, the grid, and energy storage. This involves a large number of mixed-integer variables and strong intertemporal couplings, resulting in a large-scale mixed-integer linear programming (MILP) problem. The sheer complexity of such models has become a major barrier to efficient application in real-world engineering settings. To improve solvability, decomposition strategies are often employed^[7], breaking the original problem into smaller, more tractable subproblems. These methods preserve modeling accuracy while significantly enhancing both computational efficiency and practical usability. One study [8], for example, applied Benders decomposition to solve the time-sequential production simulation model efficiently.

While decomposition strategies can break down large-scale mixed-integer programming problems into smaller subproblems to reduce complexity, several challenges persist in practice. For one, some subproblems still involve high-dimensional variables and numerous constraints, resulting in substantial computational costs per iteration. For another, the strong coupling between master and subproblems—and the limited efficiency with which cutting planes are transferred back to the master problem—can lead to slow convergence and hinder overall optimization performance. Against this backdrop, data-driven methods have emerged as a promising complement to traditional optimization techniques, particularly in the context of complex tasks like operating next-generation power systems^[9-11]. By mining and learning from the wealth of structured data generated during past solution processes, predictive models can be developed to support variable screening, constraint activation, and solution space compression. These models help reduce the scale of subproblems and accelerate convergence—improving the overall efficiency and practical applicability of the decomposition framework, without sacrificing solution quality.

This work is supported by National Natural Science Foundation Intelligent Power Grid Joint Fund Integration Project (U22B6008) .

In summary, the paper proposes an accelerated solution framework for time-sequential production simulation in integrated generation-grid-storage systems by combining Benders decomposition with data-driven techniques. The framework first establishes a model that captures the operational characteristics of generation, grid, and storage components, reflecting the system's dynamic behavior over time. Benders decomposition is then applied to decouple unit commitment from power dispatch, reducing computational complexity. To further alleviate the burden of large-scale subproblems, data-driven methods are introduced for variable screening and solution space reduction, enhancing efficiency. A case study on a provincial power system demonstrates significant improvements in solution speed while maintaining high-quality results.

The remainder of this paper is structured as follows. Section 2 introduces the foundational model for time-sequential production simulation. Section 3 presents the Benders decomposition-based optimization framework. Section 4 incorporates data-driven techniques to reduce subproblem dimensionality and improve computational performance. Section 5 concludes with the main findings and future research directions.

II. BASE MODEL FOR INTEGRATED SOURCE-GRID-STORAGE PRODUCTION SIMULATION

A. Variable Definitions

To facilitate subsequent analysis, solution procedures, and structural reformulation of the time-sequential production simulation model, the variables are categorized and defined as follows:

1) Integer Variables

The variable $u_{i,t,\text{coal}}$ represents the operating status of thermal unit i at time t ; $v_{i,t,\text{coal}}$ indicates whether the unit is starting up; and $w_{i,t,\text{coal}}$ indicates whether it is shutting down. All three are binary variables, meaning they can only take values of 0 or 1.

2) Continuous Variables

The variable $P_{i,t,\text{coal}}$ represents the active power output of thermal unit i at time t , while $R_{i,t,\text{coal}+}$ and $R_{i,t,\text{coal}-}$ indicate the upward and downward reserves it provides. $P_{i,t,\text{load}}^{\text{dc}}$ denotes the amount of load that is shed, and $P_{i,t,\text{RE}}^{\text{drop}}$ refers to curtailed wind or solar power at node i and time t . $P_{i,t,\text{RE}}$ is the total renewable output at node i , including both wind $P_{i,t,\text{wf}}$ and solar $P_{i,t,\text{pv}}$ generation. The charging or discharging power of energy storage systems is given by $P_{i,t,\text{ess}}$, while $P_{i,t,\text{load}}$ represents the demand at node i , based on an externally supplied load profile. $P_{l,t,\text{line}}$ refers to the power transmitted through line l at time t . Thermal unit ramping capabilities are captured by $D_{i,\text{coal}+}$ and $D_{i,\text{coal}-}$ indicating the maximum rates at which a unit can increase or

decrease output. Lastly, $k_{i,t,\text{ess}}^{\text{soc}}$ represents the state of charge of the storage unit at the given time and node.

B. Base Model

1) Objective Function

The objective is to minimize the total annual operating cost, expressed as:

$$\min f = F^{\text{ot}} \quad (1)$$

The annual operating cost includes the operating cost of thermal units F^{op} , the penalty for load shedding F^{dc} , and the penalty for renewable energy curtailment F^{drop} , as shown in Equation (2).

$$F^{\text{ot}} = \underbrace{\sum_{t=1}^T \sum_{i \in n_{\text{coal}}} \left((\lambda_{\text{coal}} + \lambda_{\text{co}_2})(a_i (P_{i,t,\text{coal}})^2 + b_i P_{i,t,\text{coal}} + c_i) + (c_{i,\text{coal}+} R_{i,t,\text{coal}+} + c_{i,\text{coal}-} R_{i,t,\text{coal}-}) + (c_{i,\text{coal,on}} v_{i,t,\text{coal}} + c_{i,\text{coal,off}} w_{i,t,\text{coal}}) \right)}_{F^{\text{op}}} + \underbrace{\sum_{t=1}^T \sum_{i \in n_{\text{load}}} c_{i,\text{load}}^{\text{dc}} P_{i,t,\text{load}}^{\text{dc}}}_{F^{\text{dc}}} + \underbrace{\sum_{t=1}^T \sum_{i \in n_{\text{RE}}} c_{i,\text{RE}}^{\text{drop}} P_{i,t,\text{RE}}^{\text{drop}}}_{F^{\text{drop}}} \quad (2)$$

In this equation: T is the total number of operating periods; λ_{coal} and λ_{co_2} are the unit fuel cost and carbon emission cost, respectively; a_i , b_i , c_i is the fuel consumption coefficient of thermal unit i ; $c_{i,\text{coal}+}$, $c_{i,\text{coal}-}$, $c_{i,\text{coal,on}}$, $c_{i,\text{coal,off}}$ represent the unit costs of upward reserve, downward reserve, startup, and shutdown for thermal units, respectively; $c_{i,\text{load}}^{\text{dc}}$ and $c_{i,\text{RE}}^{\text{drop}}$ are the unit penalty costs for load shedding and renewable energy curtailment.

2) Constraints

a) System Power Balance Constraint

$$\sum_{i \in n_{\text{coal}}} P_{i,t,\text{coal}} + \sum_{i \in n_{\text{RE}}} (P_{i,t,\text{RE}} - P_{i,t,\text{RE}}^{\text{drop}}) + \sum_{i \in n_{\text{ess}}} P_{i,t,\text{ess}} = \sum_{i \in n_{\text{load}}} (P_{i,t,\text{load}} - P_{i,t,\text{load}}^{\text{dc}}) \quad \forall t \quad (3)$$

b) Load Shedding Constraint

$$0 \leq P_{i,t,\text{load}}^{\text{dc}} \leq P_{i,t,\text{load}} \quad (4)$$

c) Reserve Requirement Constraint

$$\sum_{i \in n_{\text{coal}}} R_{i,t,\text{coal}+/-} \geq \sum_{i \in n_{\text{load}}} (P_{i,t,\text{load}} - P_{i,t,\text{load}}^{\text{dc}}) L^{+/-} \% + \sum_{j \in n_{\text{RE}}} P_{j,t,\text{RE}} W^{+/-} \% \quad \forall t \quad (5)$$

In this equation, $L^{+} \%$, $L^{-} \%$, $W^{+} \%$, $W^{-} \%$ represent the reserve requirement coefficients for load, wind power, solar PV, and total renewable generation, respectively.

d) Transmission Capacity Constraint

$$P_{l,\text{line}}^{\text{min}} \leq P_{l,t,\text{line}} \leq P_{l,\text{line}}^{\text{max}} \quad \forall l \in n_{\text{line}}, t \quad (6)$$

In this equation, $P_{l,\text{line}}^{\max}$ 、 $P_{l,\text{line}}^{\min}$ represent the upper and lower transmission limits of line respectively.

e) Power Flow Constraint

Based on the DC power flow model, the relationship between line flows and nodal power injections is established as follows:

$$\mathbf{P}_{t,\text{line}} = \mathbf{M}_{\text{PTDF}} \mathbf{P}_{t,\text{inj}} \quad \forall t \quad (7)$$

In this equation, $\mathbf{P}_{t,\text{line}} = [P_{l_1,t,\text{line}}, P_{l_2,t,\text{line}}, \dots]^T$ is the vector of active power flows on all lines at time t ; $\mathbf{M}_{\text{PTDF}}$ is the system's Power Transfer Distribution Factor matrix; and $\mathbf{P}_{t,\text{inj}} = [P_{i_1,t,\text{inj}}, P_{i_2,t,\text{inj}}, \dots]^T$ is the vector of active power injections at each node at time t .

f) Equipment Operational Constraints

(1) Generating Units

For thermal power units, the model primarily considers constraints on output limits, ramp rates, and minimum up/down times.

$$\begin{cases} u_{i,t,\text{coal}} P_{i,\text{coal}}^{\min} \leq P_{i,t,\text{coal}} - R_{i,t,\text{coal}} - \\ P_{i,t,\text{coal}} + R_{i,t,\text{coal}} \leq u_{i,t,\text{coal}} P_{i,\text{coal}}^{\max} \end{cases} \quad \forall i \in n_{\text{coal}}, t \quad (8)$$

$$-D_{i,\text{coal}} \leq P_{i,t+1,\text{coal}} - P_{i,t,\text{coal}} \leq D_{i,\text{coal}} \quad \forall i \in n_{\text{coal}}, t \quad (9)$$

$$\begin{cases} \sum_{\tau=T_i^{\text{ON}}+1}^t v_{i,\tau,\text{coal}} \leq u_{i,t,\text{coal}} & \forall i \in n_{\text{coal}}, t \geq T_i^{\text{ON}} \\ \sum_{\tau=t-T_i^{\text{OFF}}+1}^t w_{i,\tau,\text{coal}} \leq 1 - u_{i,t,\text{coal}} & \forall i \in n_{\text{coal}}, t \geq T_i^{\text{OFF}} \end{cases} \quad (10)$$

In this equation, $P_{i,\text{coal}}^{\min}$ 、 $P_{i,\text{coal}}^{\max}$ represent the minimum and maximum output limits of thermal unit i ; T_i^{ON} 、 T_i^{OFF} denote the minimum up time and minimum down time of the unit, respectively.

For wind and photovoltaic (PV) generation units, the model primarily considers constraints on output limits:

$$0 \leq P_{i,t,\text{RE}} \leq P_{i,t,\text{RE}}^{\max} \quad \forall i \in n_{\text{RE}}, t \quad (11)$$

In this equation, $P_{i,t,\text{RE}}^{\max}$ represents the normalized maximum output of renewable unit i at time t , determined by an externally provided time series of maximum wind or solar generation.

(2) Energy Storage Systems

The constraints for energy storage include limits on charging/discharging power and the state of charge (SOC), as well as an energy balance constraint.

$$\begin{cases} P_{i,\text{ess}}^{\text{dis},\max} \leq P_{i,t,\text{ess}} \leq P_{i,\text{ess}}^{\text{chg},\max} \\ k_{i,\text{ess}}^{\text{soc},\min} \leq k_{i,t,\text{ess}}^{\text{soc}} \leq k_{i,\text{ess}}^{\text{soc},\max} \end{cases} \quad \forall i \in n_{\text{ess}}, t \quad (12)$$

$$k_{i,t,\text{ess}}^{\text{soc}} = k_{i,t-1,\text{ess}}^{\text{soc}} + \frac{P_{i,t-1,\text{ess}}}{E_{i,\text{ess}}} \Delta T \quad \forall i \in n_{\text{ess}}, t \quad (13)$$

In this equation, $P_{i,\text{ess}}^{\text{chg},\max}$ 、 $P_{i,\text{ess}}^{\text{dis},\max}$ denote the maximum charging and discharging power of the energy storage system,

respectively; $k_{i,\text{ess}}^{\text{soc},\max}$ 、 $k_{i,\text{ess}}^{\text{soc},\min}$ represent the maximum and minimum state-of-charge ratios for the storage unit at node i ; $E_{i,\text{ess}}$ is the rated energy capacity of the storage system; and ΔT is the time step of the optimization horizon.

III. REFORMULATION OF THE TIME-SEQUENTIAL SIMULATION MODEL FOR INTEGRATED GENERATION-GRID-STORAGE SYSTEMS

A. Analysis of the Characteristics of the Time-Sequential Production Simulation Model

Given the high-dimensional, strongly coupled, mixed-integer nature of the time-sequential production simulation model for integrated generation-grid-storage systems, this paper conducts a structural analysis to explore the coupling between unit commitment variables (i.e., integer variables) and generation output variables (i.e., continuous variables). Based on this analysis, a decomposition-based optimization method is proposed, leveraging the structural separation of variable types. The coupling between these variables is primarily reflected in the following three aspects:

- Variable activation constraints: The feasibility of generation output variables depends on the activation of unit commitment variables. Specifically, output variables have physical meaning only when the unit is in the “on” state (i.e., the binary status variable equals 1), thereby forming a distinct mixed-integer coupling structure.
- Temporal dynamic characteristics: Both commitment and output variables evolve over time and exhibit intertemporal coupling. For example, unit commitment variables are constrained by minimum up/down time logic, while output variables must satisfy sequential power balance and ramping limits—further reinforcing the linkage between the two.
- Conditional decoupling: Once unit commitment decisions are fixed in a given optimization round, the associated subproblem involving generation outputs becomes a deterministic linear program with a clear structure, allowing for efficient solution.

Based on these structural insights, this paper constructs a decomposition framework that separates the unit commitment master problem from the power dispatch subproblem. The master problem focuses on optimizing combinations of binary startup/shutdown decisions, while the subproblem performs linear optimization of output variables under a fixed unit commitment schedule.

B. Reformulation of the Time-Sequential Production Simulation Model Based on Benders Decomposition

Benders decomposition, a classical master-subproblem optimization framework, achieves global convergence by decomposing large-scale mixed-integer programming problems into a master problem, a feasibility subproblem, and an optimality subproblem. Through iterative exchanges of cutting planes between the master and subproblems, the method gradually refines the solution space. In this study, the time-sequential production simulation model is reformulated using Benders decomposition by separating it into a master

problem for unit commitment decisions and a subproblem for generation dispatch optimization.

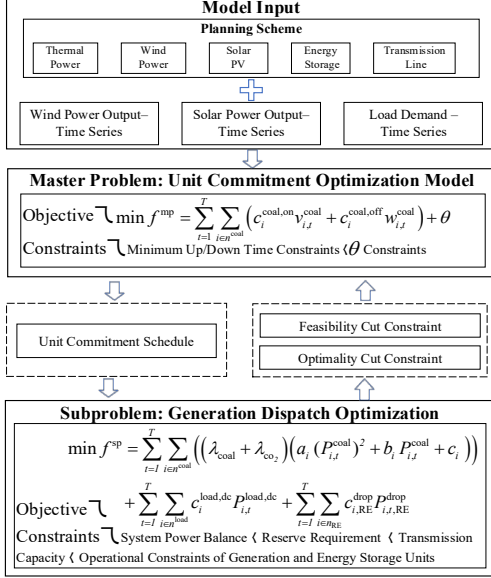


Fig. 1. Framework of the Benders-Decomposed Time-Sequential Production Simulation Model

The Benders decomposition formulation is as follows:

1) Master Problem

a) Objective Function

$$\min f^{\text{mp}} = \sum_{t=1}^T \sum_{i \in n^{\text{coal}}} \left(c_i^{\text{coal,on}} v_{i,t}^{\text{coal}} + c_i^{\text{coal,off}} w_{i,t}^{\text{coal}} \right) + \theta \quad (14)$$

In this equation, f^{mp} represents the objective function of the master problem, and θ is a continuous variable representing the lower bound of the subproblem's objective value, incorporated into the master problem.

b) Constraints

The constraints in the master problem include: the minimum up/down time constraints for thermal units (as defined in Equation (10)), as well as the cutting plane constraints introduced through Benders decomposition. These cutting planes consist of: (1) constraints representing the best-known lower bound of the objective function from previous iterations, (2) feasibility cuts, and (3) optimality cuts.

$$\begin{cases} \theta \geq \theta_0 \\ 0 \geq \partial^T(\mathbf{b} - \mathbf{B}\mathbf{v}) \\ \theta \geq \partial^T(\mathbf{b} - \mathbf{B}\mathbf{v}) \end{cases} \quad (15)$$

In this equation: θ_0 is the current best estimate of the lower bound of the objective function, used to guide $\partial^T(\mathbf{b} - \mathbf{B}\mathbf{v})$ denotes the cutting plane expression derived from the dual solution of the subproblem, used to construct both feasibility and optimality cuts; $\mathbf{v}$ represents the vector of integer variables shared between the master and subproblem—in this case, the unit commitment status variables; $\mathbf{B}$ is the coefficient matrix that maps the coupling variables $\mathbf{v}$ into the structural constraints of the subproblem; and $\mathbf{b}$ is the right-hand side vector of the subproblem constraints.

2) Subproblem

a) Objective Function

$$\begin{aligned} \min f^{\text{sp}} = & \sum_{t=1}^T \sum_{i \in n^{\text{coal}}} \left((\lambda_{\text{coal}} + \lambda_{\text{co}_2}) (a_i (P_{i,t}^{\text{coal}})^2 + b_i P_{i,t}^{\text{coal}} + c_i) \right) \\ & + \underbrace{\sum_{t=1}^T \sum_{i \in n^{\text{load}}} c_i^{\text{load,dc}} P_{i,t}^{\text{load,dc}}}_{F_{\text{dc}}} + \underbrace{\sum_{t=1}^T \sum_{i \in n_{\text{RE}}} c_i^{\text{drop}} P_{i,t}^{\text{drop}}}_{F^{\text{drop}}} \end{aligned} \quad (16)$$

In this equation, f^{sp} represents the objective function of the subproblem.

b) Constraints

The subproblem includes the following constraints: system active power balance, load shedding requirements, reserve requirements, transmission capacity limits, generation output limits, and operational constraints of energy storage systems—corresponding to Equations (3) through (9), and (11) through (13) as previously defined.

Given the unit commitment schedule determined by the master problem, the subproblem optimizes the continuous variables subject to these linear constraints. To enable iterative interaction with the master problem, the subproblem is reformulated in its dual form. Based on the dual solution, feasibility cuts or optimality cuts (as described earlier $\partial^T(\mathbf{b} - \mathbf{B}\mathbf{v})$) can be generated and fed back into the master problem. By iteratively refining the solution space, this process ensures convergence of the overall model.

3) Convergence Criterion

Convergence is considered achieved when the objective values of the master problem and the subproblem satisfy the following predefined tolerance condition:

$$|f^{\text{mp}} - f^{\text{sp}}| < \varepsilon \quad (17)$$

In this equation, ε denotes the predefined convergence tolerance.

When this condition is satisfied, it indicates that the solutions of the master and subproblems have approached global optimality, and the difference in objective values falls within an acceptable range. At this point, the Benders decomposition iteration is terminated.

IV. ACCELERATED SOLUTION OF THE DECOMPOSED TIME-SEQUENTIAL SIMULATION MODEL VIA GAUSSIAN PROCESSES

A. Gaussian Process Classifier: Modeling and Training

As 0–1 integer variables, unit commitment decisions (i.e., on/off status) can cause a combinatorial explosion in problem size within large-scale mixed-integer linear programming models. Traditional methods such as branch-and-bound often become computational bottlenecks when handling such variables. To address this challenge, this study introduces a predictive model for unit commitment status, trained on historical solution samples, with the goal of pre-screening feasible value spaces and guiding the optimization process probabilistically. By fixing part of the binary variables with high-confidence predictions prior to solving the master problem, computational efficiency can be significantly improved.

Gaussian processes offer strong capabilities in small-sample learning and uncertainty quantification, making them well-suited for classification tasks in complex settings. To enable pre-screening of unit commitment status variables for thermal power units, this study develops a data-driven model based on a Gaussian Process Classifier (GPC). The model is trained on historical solution data accumulated during Benders decomposition iterations, leveraging the strengths of small-sample learning while providing confidence-aware predictions. This allows for both status forecasting and reliability assessment of the predicted outcomes. The overall modeling and training process is outlined as follows:

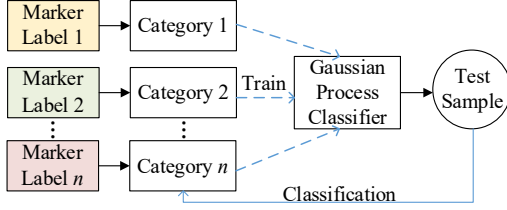


Fig. 2. Schematic of the Data-Driven Approach

- Step 1: Construction of Training Samples

This study builds the training dataset based on feasible solutions recorded during the Benders decomposition process, extracting their temporal features for model input. Each input feature vector $\mathbf{X}$ consists of: [current time index, unit output value, output as a percentage of unit capacity, output as a percentage of total system output, product of output share and total system load shedding, startup indicator, shutdown indicator]. The corresponding output label $\mathbf{Y}$ is the unit's on/off status at the same time step in a future iteration, where 1 indicates startup and 0 indicates shutdown. The initial dataset contains 400 samples, and newly generated data from each iteration are continuously added to the historical dataset to ensure broad coverage of representative operating conditions and system states.

- Step 2: Kernel Selection and Model Specification

The Gaussian Process model in this study is constructed using the commonly adopted squared exponential (SE) kernel. The covariance function is defined as follows:

$$k(\mathbf{x}_i, \mathbf{x}_j) = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2l^2}\right) \quad (18)$$

In this equation, σ_f denotes the signal variance (output scale), and l is the characteristic length scale. Together, they form the set of hyperparameters $\theta = \{\sigma_f, l\}$ for the kernel function.

- Step 3: Classifier Training

This study employs a maximum marginal likelihood approach to infer the hyperparameters of the Gaussian Process. Specifically, the hyperparameters are learned by maximizing the log marginal likelihood function:

$$\ln P(\mathbf{Y}|\mathbf{X}, \theta) = -\frac{1}{2} \mathbf{Y}^T \mathbf{K}^{-1} \mathbf{Y} - \frac{1}{2} \ln |\mathbf{K}| - \frac{n}{2} \ln 2\pi \quad (19)$$

In this equation, $\mathbf{K}$ is the covariance matrix of the training samples, computed as $\mathbf{K} = k(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I}_n$; σ_n^2 represents the noise variance, and $\mathbf{I}_n$ is the identity matrix.

The training process iteratively solves for the hyperparameters θ using gradient-based optimization, ultimately determining the optimal kernel function structure.

- Step 4: Prediction and Pre-screening Application

After completing offline training, the Gaussian Process Classifier is used to predict the posterior probability distribution of unit commitment status for each variable under consideration. For each candidate variable, its corresponding feature vector is fed into the trained model to obtain a probabilistic prediction of the on/off state. Based on the predicted confidence level, a pre-screening rule is applied: if the predicted probability exceeds a specified threshold (or falls below $1 - \text{threshold}$), the variable is classified as a deterministic state and fixed accordingly. Variables falling within the uncertain range are retained as optimization variables and continue to participate in the subsequent optimization process.

B. Pre-screening of Variables and Constraints Based on Gaussian Processes

Based on the prediction results from the Gaussian Process Classifier, this study proposes a pre-screening strategy targeting both unit commitment variables and redundant constraints in the subproblem. The specific procedures are outlined below.

1) Pre-screening of Unit Commitment Variables

Using the confidence scores output by the Gaussian Process Classifier, unit commitment variables are divided into two categories: deterministic and uncertain. A tiered handling strategy is then applied accordingly:

- Pre-fixing Deterministic Commitment States

The classifier is used to predict the on/off status of thermal units. For those variables with predicted probabilities exceeding a predefined confidence threshold, their commitment status is treated as deterministic and fixed in advance. These fixed variables are incorporated as constants in the subsequent optimization model, thereby reducing the discrete decision space of the master problem.

- Probability-Guided Screening of Uncertain States

For variables falling within the intermediate probability range—i.e., not confidently classified as on or off—a dynamic weighting coefficient

ℓ_i is constructed based on the predicted uncertainty. This coefficient is designed as a monotonically increasing function of prediction uncertainty and is added to the objective function as a penalty term. By assigning a higher optimization cost to commitment decisions with greater uncertainty, the solver is guided to prioritize candidate solutions with higher overall prediction confidence within the feasible region.

2) Pre-screening of Redundant Subproblem Constraints

For units already determined to be offline, their power output variables are mathematically constrained to zero. This allows identification and removal of related constraints that no longer influence the feasible region—such as ramp rate limits or line flow constraints involving those units. These constraints, while present in the model, do not affect the solution and only increase the model's dimensionality and computational burden. By systematically identifying and

eliminating such redundant constraints—especially when combined with sparse matrix representations—the model's complexity can be significantly reduced.

V.CASE STUDY

To verify the effectiveness of the proposed method, a simulation study was conducted using a provincial-level power system as a case example. The network topology of the system is shown in Figure 3, consisting of 90 nodes and 200 transmission lines. The system's initial configuration is as follows: thermal power capacity of 83.2 GW, wind power capacity of 13 GW, solar PV capacity of 18 GW, energy storage capacity of 2 GW, and a maximum load capacity of 100 GW.

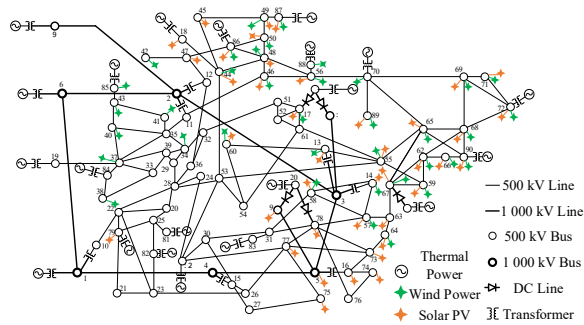


Fig. 3. Topology diagram of a provincial case system

In this study, the baseline method (hereafter referred to as the original method) is defined as the solver's default integrated optimization approach. The proposed method, which combines Benders decomposition with data-driven techniques, is used as the improved solution strategy. Simulations are conducted over three different durations—4 days, 7 days, and 30 days—with a time step of 1 hour, resulting in an equivalent daily simulation horizon of 24 hours. Under these varying time scales, the computational time and optimal objective values of the two methods are compared, with results summarized in Tables 1 and 2. As shown in the results:

TABLE I COMPUTATIONAL TIME COMPARISON

Solution Method	4-Day Computation Time	7-Day Computation Time	30-Day Computation Time
Original Method	23.12 s	221.89 s	3917.56 s
Improved Method	9.70 s	60.82s	807.23 s

TABLE II OPTIMAL OBJECTIVE VALUE COMPARISON

Solution Method	4-Day Optimal Objective Value	7-Day Optimal Objective Value	30-Day Optimal Objective Value
Original Method	680 million	985 million	3162 million
Improved Method	684 million	1001 million	3254 million

Compared with the original method, the proposed improvement achieved 38.5%, 27.41%, and 20.6% of the

original computation time for simulation durations of 4 days, 7 days, and 30 days, respectively. Meanwhile, the relative differences in the optimal objective values were kept within 0.6%, 1.6%, and 2.9%. These results indicate that, under an acceptable level of accuracy loss, the improved method significantly enhances computational efficiency, demonstrating both strong numerical performance and practical engineering applicability.

VI.Conclusion

The proposed time-sequential simulation approach—integrating Benders decomposition with data-driven techniques—effectively addresses the challenges posed by the high-dimensional and strongly coupled structure of the generation-grid-storage coordination model. Validation using a case study based on a provincial-level power system demonstrates the method's effectiveness in improving computational efficiency. This approach offers a new pathway for the practical application of time-sequential production simulation in engineering contexts and provides a theoretical foundation for power and energy balance risk assessment in next-generation power systems.

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